

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	25 June 2026
Team ID	SWTID-2026-4219
Project Name	Visualization Tool for Electric Vehicle Charge and Range Analysis
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Collection	Import Electric Vehicle datasets (EV data, Charging Stations, EV India, Cheapest EV Cars).
FR-2	Data Processing	Clean, validate, and organize datasets for analysis.
FR-3	SQL Analysis	Execute SQL queries to analyze vehicle range, pricing, charging stations, brands, and efficiency.
FR-4	Dashboard Visualization	Create interactive Tableau dashboards with charts, maps, filters, and KPIs.
FR-5	Story Visualization	Develop Tableau Story to present insights in a sequential format.
FR-6	Dashboard Integration	Embed Tableau Dashboard and Story into the Flask web application.
FR-7	User Interaction	Allow users to explore dashboards using filters, maps, and interactive visualizations.
FR-8	Project Repository	Store source code, datasets, SQL scripts, documentation, and project files in GitHub.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	User-friendly interface with interactive dashboards and easy navigation.
NFR-2	Performance	Fast loading and smooth visualization of EV datasets.
NFR-3	Reliability	Accurate data processing and consistent dashboard performance.
NFR-4	Availability	Dashboard and project accessible through Tableau Public and GitHub.
NFR-5	Scalability	Architecture supports adding new datasets, dashboards, and analytical features.
NFR-6	Maintainability	Modular project structure for easy updates and future enhancements.
NFR-7	Security	Secure project hosting using GitHub and HTTPS-enabled Tableau Public links.